Group 4 – Module 08 Lab Reflection
Course: ITAI 1371 – Introduction to Machine Learning
Lab: Module 08 – Bias-Variance Tradeoff and Overfitting
Group: Group 4

Member 1 – Patrick Rostand Gandjouon Tchassem

My Reflection

This lab helped me understand the difference between underfitting and overfitting better. Before working on the lab, I knew that a machine learning model could be too simple or too complicated, but I did not fully understand how that affects predictions. After running the code and looking at the graphs, it started to make more sense.

One thing I found interesting was comparing the three polynomial models. The degree 1 model was just a straight line and could not follow the pattern in the data. The degree 4 model followed the pattern better, while the degree 15 model was more complicated and started following small changes in the data. This helped me see that making a model more complex does not always mean it will perform better.

The learning curves were another important part of the lab for me. I noticed that the degree 1 model had low scores because it was too simple to learn the pattern. The degree 15 model had a big difference between its training and validation scores. Some of the validation scores were extremely low, which showed me that a model can work well with training data but struggle when tested on different data.

I also learned that we cannot judge a model only by how well it performs during training. We need to check how it performs on data it has not seen before. Overall, this lab gave me a better understanding of bias, variance, and why finding the right model complexity is important in machine learning.

Member 2 – saimi manasiya
In this lab, I learned about the bias-variance tradeoff and how model complexity affects the performance of machine learning models. I understood the difference between underfitting and overfitting by comparing polynomial regression models with degrees 1, 4, and 15. The degree 1 model was too simple to capture the pattern in the data, while the degree 4 model provided a better fit. The degree 15 model was more complex and showed signs of overfitting because it performed better on training data but struggled to generalize to unseen data.

I also learned how learning curves help us understand a model’s performance by comparing training scores and cross-validation scores as the number of training examples increases. The degree 1 model showed signs of underfitting because it struggled to learn the underlying pattern even with more training data. The degree 15 model showed a larger difference between training and cross-validation performance, indicating high variance and poor generalization. This activity helped me understand why choosing the right model complexity is important in machine learning.

Overall, this lab improved my understanding of polynomial regression, learning curves, underfitting, overfitting, and generalization. I learned that a complex model is not always the best model and that the goal is to find a balance between bias and variance. These concepts will help me understand how to build machine learning models that perform well on both training data and new, unseen data.

Member 3 – Kenneth Kouokam

For Module 08, my assigned role is Synthetic Data and Initial Exploration. The main idea I take from this setup is that understanding the data comes before deciding how complicated a model should be. The lab uses a sine function with added random noise, which makes the distinction between the underlying relationship and individual observations easier to understand. The target is numerical, so

this is a regression problem even though the pattern is curved.

The added noise gives the exercise a useful purpose. Without it, the observations would sit exactly on the sine curve. With it, individual points can be above or below the underlying pattern. This helps
me understand why matching every training point is not necessarily the goal. A very flexible model might follow random fluctuations that will not appear in another sample. I can connect this to overfitting: a model can look successful on the data it has already seen while being less reliable on new examples.

The same example also helps explain underfitting. A straight line cannot represent all of the changes in direction across this sine-wave pattern. However, simply choosing the most complex model is not a complete solution either. The initial scatter plot provides context for comparing the later curves, but it
cannot establish how well those models generalize. That is why the learning curves and validation scores in the other sections are important. They provide evidence beyond how closely a fitted line follows the displayed points.

Reproducibility is another useful part of the setup. Setting the random seed to 0 makes it possible to
generate the same noise again when the code is rerun in the same way. This matters in a group project because changing the random observations while comparing models could make the differences harder to interpret. A fixed seed does not make the data better or guarantee good performance. It helps us keep the starting conditions consistent so we can focus on model complexity.

The shape of the data also has a practical meaning. X needs 100 rows and one feature column, while y contains one target value for each observation. Reshaping X changes how the inputs are organized for the model; it does not create new observations. My takeaway from this role is that setup choices support the whole experiment. A clear plot, correctly organized arrays, and repeatable data help the group interpret the later results. A question I would like to explore is whether the conclusions remain
similar when the noise level or random seed changes.

Member 4 – Anavictoria Cavazos

After working through the google colab for lab 08 I learned about the importance of having a complexity balance in the model.
This was seen when the polynomial degrees 1, 4, and 15 were used, 1 was underfitting, 15 was overfitting, but 4 was perfect, sort
of reminds me of the famous children's story, "Goldilocks and the Three Bears". Although that has nothing to do with machine learning it
helps understand the concept of finding the right polynomial degree for a model when working with polynomial regression.

Another aspect I will point out when working through the colab was the concept of the visual aids such as the graphs, shown in task 1 and
in task 2. The visual displayed in task 1 showed the comparison of all three polynomials with data points then the two graphs for task 2
show the testing score with the cross validation. Now when I was first going through the task and then the idea of learning curve I was
not fully understanding how it all worked, but when I looked at the lines for all three degrees it all made sense how they all interlink.
Based on my understanding degree 1 is terrible because it is literally a staright line, it has no where to move where it can cath data,
which is why it is 'under' fitting. Then the line for degree 15 is way all over the place that it catches some of the noise, which is why
it performs poorly on new unseen data. This idea shown with the lines and the data points is then displayed on the learning curves to
help see why and where the mistake occurs and how that looks when comparing the training and cross validation scores.

Overall, I really enjoyed working on lab 08 because the idea of underfitting and overfitting a model is very important, but sometimes can
be seen as a simple concept. The risk of viewing this lesson in that manner is that one will not take this into consideration when
preparing their model, but when one takes into mind these ideas it helps the flow of the model to be much smoother and better.

Member 5 – Hashim Sayed Hoosini

When I first started learning about machine learning, I used to think that the biggest threat to a model was overfitting—making something too complicated that memorizes random noise. This lab shifted my focus entirely to the opposite problem, which is underfitting, or having high bias. I realized that underfitting is like trying to draw a circle using only straight, rigid lines. No matter how hard you try, a simple tool cannot capture a curved reality. In our exercise, forcing the Degree 1 linear model onto a wavy sine curve showed me that a model making overly simple assumptions is doomed from the start because it lacks the basic flexibility to see the true pattern.

What made this concept clear to me was watching the learning curves develop for these two extreme models. I always assumed that if a model was performing poorly, the easiest fix was to simply feed it more data. However, the flat lines on the learning curve for our simple Degree 1 model completely disproved this myth. Both the training and the testing lines flattened out almost immediately, staying very close together at a terrible score. This visual taught me that when a model has high bias, adding more training examples is like trying to learn to play the piano by reading a math textbook over and over. No matter how much data you throw at it, the Degree 1 model cannot improve because its underlying structure is fundamentally incapable of learning the pattern.

This behavior is completely different from the overfit Degree 15 model, where the learning curve shows a massive gap between excellent training scores and terrible validation scores. With high bias in the Degree 1 model, there is almost no gap at all because the model is equally bad at predicting both the data it has already seen and the new data it is tested on; the curves quickly run parallel to each other in a flat line of low performance. Meanwhile, the Degree 15 model tries to chase every tiny, random bump in the data, which gives it a deceptively perfect score during training but causes its performance to crash when faced with anything new. This taught me that the "gap" on a learning curve is a direct measurement of a model's struggle. A tight, low-performing gap (Degree 1) means the model is just too stubborn to learn, while a huge gap (Degree 15) means the model is too hyper-focused on tiny details.

Ultimately, this lab changed how I think about building models. It showed me that finding the right balance is a delicate process of negotiation. We cannot just default to the simplest model out of fear of overfitting, because we risk running into a wall where our model is completely blind to the underlying trend. Moving forward, when I look at a learning curve and see those two lines flatlining together, I will know immediately that it is time to stop looking for more data and instead start building a more flexible, creative model that can actually match the complexity of the real world.